

Evaluation of six commercial test strips for measuring nitrate in cattle drinking water

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Abstract

Nitrate toxicosis from contaminated drinking water is a common cause of acute death in cattle. The objective of this study was to determine if commercially available semi-quantitative colorimetric test strips designed for human drinking water can be used as a screening test to identify elevated nitrate concentrations in cattle drinking water. Six test strip brands were used to measure nitrate concentrations in water samples with known values. Sample concentrations were prepared and verified via ion chromatography. Ten volunteers were instructed to test water samples and record nitrate levels according to the test strip manufacturer's instructions. There was no significant effect of evaluator or evaluator sex. Results were dichotomized where targeted nitrate concentrations ≥ 100 ppm were considered positive and < 100 ppm negative, resulting in a 50% proportion of positive samples. Sensitivities of the tests ranged from 22.5% to 97.5%. Two brands resulted in few false negatives when screening water samples with sensitivities of 97.5% and 90%, two brands resulted in a moderate number of false negatives with sensitivities of 77.5% and 72.5%, and two brands resulted in numerous false negatives with sensitivities of 47.5% and 22.5%. These findings suggest that while some commercially available test strips can be an effective and economical option for producers and veterinarians to utilize for screening water sources for nitrate contamination, others may pose a risk of underestimating harmful nitrate levels. Water samples with nitrate concentrations reaching or exceeding 100 ppm on test strips should undergo confirmatory testing via quantitative analysis.

Key words: cattle, nitrate toxicosis, drinking water, colorimetric strip testing, screening test

Introduction

Nitrate toxicosis has been observed in ruminant species, specifically cattle, for many years.¹ A wide variety of animal species are susceptible to nitrate poisoning; however, cattle are most frequently affected.² Nitrate toxicosis is a common cause of acute death loss of cattle in the United States. A retrospective study in California found that nitrate/nitrite poisoning was the most commonly diagnosed toxicosis of cattle.³ In cattle, acute toxicosis occurs following the reduction of nitrate to nitrite by microflora within the rumen, with nitrite being approximately 10 times more toxic than nitrate. The nitrite

ions are absorbed into the bloodstream where they combine with red blood cells and oxidize hemoglobin into methemoglobin.² The heme iron in hemoglobin is normally present in the reduced valence form, but when the heme iron group is exposed to nitrite, it becomes oxidized to methemoglobin and is incapable of oxygen transport.⁴ When methemoglobin concentrations exceed 30 to 40%, tissue anoxia occurs. Clinical signs associated with nitrate toxicosis appear rapidly due to tissue hypoxia and hypotension, and include rapid, weak heartbeat, decreased body temperature, muscle tremors, weakness and ataxia. When the methemoglobin concentration reaches harmful levels (30 to 40% for clinical signs, 70 to 80% for death), brown, cyanotic mucous membranes may be observed. Animals that have died from acute nitrate toxicosis are typically identified postmortem due to the presence of characteristic chocolate-brown colored blood and tissues.²

Ingestion of nitrates from forages, specifically nitrate-accumulating grasses, sorghums and weeds, and contaminated water supplies can have an additive effect resulting in acute toxicosis.⁵ Forages normally contain a small amount of nitrate, but under certain conditions (e.g., drought, excessive fertilization), nitrate can accumulate to toxic concentrations.⁶ There are also several animal-level factors that influence an animal's susceptibility to nitrate toxicosis in ruminants including ruminal microflora composition, current health and pregnancy status, age, and diet composition. Water may be contaminated with toxic levels of nitrate from fertilizer, animal waste or decaying organic matter.⁶ While nitrogen-based fertilizers are an important component of agriculture, their use presents potential opportunities for toxic exposure to livestock. Direct consumption of fertilizers or contaminated drinking water are routes by which livestock can be exposed to harmful levels of nitrates. Although it is not recommended, during shortage situations, drinking water has reportedly been hauled to livestock using containers previously used to transport fertilizer.^{7,8} This poses a serious risk of poisoning from nitrate or other dangerous fertilizer components.

In February 2025, the Kansas Veterinary Diagnostic Laboratory (KVDL) was referred a case of suspected nitrate toxicosis in a herd of cattle located in Kansas. The referring veterinarian reported that 89 of 150 cattle had died following exposure to severe cold weather while grazing on milo stalks for 9 days. Notably, the case history included water being hauled to the cattle in an old fertilizer tank. Diagnostic samples submitted

included ocular fluid from 5 deceased cows and 2 samples of accessible drinking water. All 5 ocular fluid samples had elevated nitrate levels ranging from 25 to 250 parts per million (ppm). Since ocular fluid nitrate greater than 20 ppm is suggestive of nitrate toxicosis, it was surmised these cattle likely died from exposure to elevated nitrate levels. The water samples were analyzed using ion chromatography^a, yielding nitrate concentrations of 9,416 and 9,595 ppm. Water nitrate concentrations greater than 100 ppm should stimulate quantitative analysis of other dietary sources, since water and dietary nitrate are additive. Based on these findings, the case toxicologist concluded that the severely elevated nitrate concentration in the water was the most likely cause of death.

Consultation on this case prompted the toxicology service to consider approaches to on-farm assessment of nitrate concentrations in drinking water. While reliable and validated field tests are available to determine the safety of nitrate levels in forages, on-farm tests for water sources are lacking. Commercially available test strips, which allow for semi-quantitative estimation of water contaminants, including nitrate, have been utilized to screen sources of human drinking water, but have not been evaluated for sources of livestock drinking water.⁹ These tests are simple to use, quick and inexpensive. Screening livestock water sources using nitrate test strips could be an affordable and convenient way to avoid the disastrous consequences of nitrate poisoning. By comparing commercially available colorimetric strip tests to gold standard analytical techniques, we aimed to assess whether these tests could meet this need.

The objective of this study was to determine whether commercially available semi-quantitative colorimetric test strips designed for human drinking water can serve as a high-sensitivity screening tool for detecting nitrate concentrations associated with toxicosis risk in cattle, and to evaluate their diagnostic performance at nitrate concentrations exceeding human safety thresholds.

Materials and methods

Selection of index tests

To be eligible for selection as an included index test, colorimetric test strips needed to be commercially available (as of May 2025), be marketed for drinking water, and have nitrate test pads with a test key capable of measuring concentrations up to 500 ppm. Test strips marketed for testing nitrate concentrations in swimming pools and those with concentration keys unable to detect nitrate concentrations up to 500 ppm were excluded from selection eligibility. Five of the 6 evaluated brands of test strips were identified through a search of and sourced from Amazon (amazon.com). The SJ Wave[®] 16-in-1 Water Test Kit^b and Varify[®] 17-in-1 Complete Drinking Water Test Kit^c were selected because their test key color gradients were darker or lighter, respectively, compared to other eligible tests. The other 3 selected tests were chosen based on their relatively high (Bartovation[®] Nitrite, Nitrate Two Pad Test Strip^d), medium (JNW Direct[®] Drinking Water Test Kit 15 in 1^e), or low (AssuTest[®] 17-in-1 drinking water test kits^f) prices. An additional test (MilliporeSigma[™] MQuant[™] Nitrate Test Strips^g) was selected based on its availability from Fisher Scientific (fishersci.com), a popular laboratory supply company. Index test information is summarized in Table 1.

Table 1: Overview of nitrate test strips: Brand, cost and basis for study inclusion.

Brand name	Product name (number of test strips)	Total test price* (Price/strip) (USD)	Reason for test selection
^b SJ wave [®]	16-in-1 Water Test Kit (100 strips)	\$16.99 (\$0.17)	Color key scale appeared to have a darker gradient compared to other available test options.
^c Varify [®]	17-in-1 Complete Drinking Water Test Kit (100 strips)	\$26.95 (\$0.27)	Color key scale appeared to have a lighter gradient compared to other available test options.
^d Bartovation [®]	Nitrite, Nitrate Two Pad Test Strip (50 strips)	\$24.22 (\$0.48)	High per unit price point.
^e JNW Direct [®]	Drinking Water Test Kit 15 in 1 (100 strips)	\$11.99 (\$0.12)	Mid-range per unit price point.
^f AssuTest [®]	17-in-1 drinking water test kits (125 strips)	\$8.99 (\$0.07)	Low per unit price point.
^g Mquant [™]	Nitrate Test Strips, colorimetric, 10-500 mg/L (NO3-) (100 strips)	\$98.90 (\$0.99)	Test strips available from a laboratory supply company.

* Prices at the time test strips were ordered in May 2025.
^b SJ wave[®] 16-in-1 Water Test Kit, SJ Wave, Mount Waverley, VIC, Australia
^c Varify[®] 17-in-1 Complete Drinking Water Test Kit, Varify Inc., Coronado, CA
^d Bartovation[®] Nitrite, Nitrate Two Pad Test Strip, Bartovation, West Harrison, NY
^e JNW Direct[®] Drinking Water Test Kit 15 in 1, JNW Direct, Pinner, GB
^f AssuTest[®] 17-in-1 drinking water test kits, Hangzhou, China
^g MilliporeSigma[™] MQuant[™] Nitrate Test Strips, Burlington, MA

Preparation of reference standards

Reference standard dilutions were prepared by a single author (BAH) and a laboratory technician using a 1,000 ppm nitrate solution^h. The 1,000 ppm nitrate solution was diluted using 18 Ω water to the following target concentrations: 500 ppm, 250 ppm, 100 ppm, 75 ppm, 50 ppm and 0 ppm, with the 100 ppm and 50 ppm samples being duplicated, resulting in a total of 8 water samples at each test station. Following preparation, the concentrations were verified using the KVDL validated diagnostic protocol for the detection of nitrate in water samples via ion chromatography^a, yielding test solutions of 492 ppm, 234 ppm, 87 ppm, 67 ppm, 44 ppm and 0 ppm. Ion chromatography was selected as the reference test because it can separate nitrate ions from other anions via charge and can quantify values by measuring conductivity and comparing to a known standard of nitrate.

Data collection

Data was collected in a prospective manner over 2 days utilizing data collection sheets designed by the authors (Appendix A). Ten volunteers with no prior experience with the test strips and no known color blindness evaluated the index tests as evaluators. The evaluators were blinded to test brand and reference standard concentration. They used the 6 test strip brands to test and record nitrate levels in the supplied reference standard dilution water samples according to each test manufacturer's instructions. Evaluators (5 males and 5 females) were randomly assigned an identification number. Test strip brands were randomly assigned an identification letter, and dilution samples were randomly assigned an order using identification letters within the test stations, all using a random number generator in Excelⁱ (RAND function). Evaluators were also assigned a random order to rotate through the testing stations.

Stations were set up for each test strip brand, with the test strips in their original packaging after all identifying information had been hidden, leaving only the instructions and test key visible to evaluators. Reference standard dilution water samples were also placed at each station (0 ppm (18 Ω water), 50 ppm (n = 2), 75 ppm, 100 ppm (n = 2), 250 ppm, and 500 ppm), arranged in the randomized order within each test station.

Evaluators were supplied with paper copies of data collection sheets arranged in randomized order in which they were to rotate through the test stations. At each test station, evaluators were instructed to prepare one test strip for each sample (8 samples per test station), remove the cap of each sample dilution, taking care to keep the dilutions in the appropriate order, and proceed with testing the sample dilutions according to instructions supplied by the test manufacturer. Authors (BAH, CJH, SAF) oversaw data collection by evaluators and acted as a timer to standardize the timing of reading test strip results. The designated timer announced when 30 seconds and 60 seconds had passed since the index tests had varied recommended intervals regarding when test results should be read. Numerical values of nitrate concentrations (ppm) for each dilution were recorded on the provided data collection forms. After the evaluators completed testing the samples at their first station, they moved to the next station and repeated the same procedure until all samples at each station had been tested.

Statistical analysis

Data from the collection sheets were manually entered into an Excelⁱ spreadsheet and then were analyzed in STATA 16.^j and R^k using RStudio^l by a single author (MWS) who was not

involved with reference standard preparation and was blinded to index test brand and evaluator identity. Targeted sample concentrations for the reference standard and test strip values were dichotomized; ≥ 100 ppm nitrate was considered a positive sample, and < 100 ppm a negative sample. Cutoff points for positive and negative test results were pre-specified based on historical nitrate toxicosis cases and diagnostic experience of the authors. The prevalence of positive samples based on the dichotomization of reference standards at 100 ppm resulted in a 50% proportion of positive samples (4 per test station). Subsequent analysis was of test strip agreement with the reference standard category (positive or negative). The effect of sex and evaluator on test agreement was assessed using a logistic regression of agreement in STATA. Individual test strip performance was analyzed for sensitivity, specificity, false negative risk (at 50% prevalence), and likelihood ratios using the R package *epiR*. Additionally, test agreement with the reference standard values was summarized for major differences where the reference standard target concentration was ≥ 250 ppm and the test result was < 100 ppm. No intermediate or missing index test or reference standard test results occurred or needed to be handled. Because this was a small proof of concept-type project, a sample size justification was not performed. The number of evaluators selected was based on volunteer availability during the project timeframe (June 2025). The number of index tests selected for evaluation was chosen based on the number of commercially available options and budget constraints. The number and concentration of water samples tested were chosen to correspond with the color scales of the index tests and to include biologically relevant thresholds to meet the study objective.

Results

There was no significant effect of evaluator ($P = 1.0$) or evaluator sex ($P = 0.656$). Test agreement results are summarized for all tests in Table 2. With the cutoff point set at 100 ppm, 2 tests had sensitivities of $\geq 90\%$, (Bartovation and Mquant), 2 tests had sensitivities of approximately 70%, (SJ wave and AssuTest), and 2 tests had sensitivities less than 50% (JNW Direct and Varify). The likelihood ratio for a positive test was the largest (best) for SJ wave, and the likelihood ratio for a negative test was the smallest (best) for Bartovation. SJ wave and Bartovation had no major differences in agreement with the reference standards, AssuTest had one major disagreement, Mquant and JNW Direct had 2 major disagreements, and Varify had 13 major disagreements.

Discussion

Cattle can be exposed to nitrates through multiple routes, and multiple factors can affect an individual animal's susceptibility or tolerance, including health status and diet composition.² Therefore, total dietary intake of nitrates/nitrites is considered during diagnostic workups of cases. The acute toxic dose of nitrate in ruminants is likely around 500 mg/kg body weight, but can vary widely based on acclimatization.¹⁻² Forage concentrations more than 10,000 ppm (1%) dry weight can cause acute death, but typically nitrate exposure from water sources is more concerning since nitrate in water is more bio-available, whereas nitrate must leach out of forages first before being converted to nitrite.¹⁰ Water nitrate levels above 100 ppm can contribute significant amounts of nitrates to the diet, and concentrations $> 1,000$ ppm can cause acute death.

Table 2: Diagnostic performance of test strips compared to ion chromatography when utilizing a concentration of ≥ 100 ppm nitrate as a threshold for a positive test. Results show sensitivity, specificity, predictive values and likelihood ratios (with 95% confidence intervals) for each test strip brand based on the known 50% prevalence of positive samples (4 true positives and 4 true negatives).

Test	Sensitivity (95% CI)	Specificity (95% CI)	Positive predictive value (95% CI)	Negative predictive value (95% CI)	Likelihood ratio for a positive test (95% CI)	Likelihood ratio for a negative test (95% CI)
^d Bartovation®	97.5% (94.1, 100.9%)	72.5% (62.7, 82.3%)	78.0% (68.9, 87.1%)	96.7% (92.7, 100.0%)	3.55 (2.14, 5.88)	0.03 (0.01, 0.24)
^g Mquant™	90.0% (83.4, 96.6%)	60.0% (49.3, 70.7%)	69.2% (59.1, 79.3%)	85.7% (78.1, 93.4%)	2.25 (1.52, 3.33)	0.17 (0.06, 0.44)
^b SJ wave®	77.5% (68.4, 86.7%)	97.5% (94.1, 100.0%)	96.9% (93.1, 100.0%)	81.3% (72.7, 89.8%)	31 (4.44, 216.26)	0.23 (0.13, 0.41)
^f AssuTest®	72.5% (62.7, 82.3%)	95.0% (90.2, 99.8%)	93.6% (88.2, 98.9%)	77.6% (68.4, 86.7%)	14.5 (3.71, 56.73)	0.29 (0.17, 0.48)
^e JNW Direct®	47.5% (36.6, 58.4%)	100.0% (100.0, 100.0%)	100.0% (100.0, 100.0%)	65.6% (55.2, 76.0%)	∞	0.53 (0.39, 0.71)
^c Varify®	22.5% (13.4, 31.7%)	97.5% (94.1, 100.0%)	90.0% (83.4, 96.6%)	55.7% (44.8, 66.6%)	9.0 (1.20, 67.78)	0.80 (0.67, 0.95)

^b SJ wave® 16-in-1 Water Test Kit, SJ Wave, Mount Waverley, VIC, Australia

^c Varify® 17-in-1 Complete Drinking Water Test Kit, Varify Inc., Coronado, CA

^d Bartovation® Nitrite, Nitrate Two Pad Test Strip, Bartovation, West Harrison, NY

^e JNW Direct® Drinking Water Test Kit 15 in 1, JNW Direct, Pinner, GB

^f AssuTest® 17-in-1 drinking water test kits, Hangzhou, China

^g MilliporeSigma™ MQuant™ Nitrate Test Strips, Burlington, MA

Several reports have documented the catastrophic results of cattle consuming nitrate-contaminated water. Cattle deaths have been reported following water consumption from nitrate-contaminated wells,¹¹⁻¹² piped water,¹³ and ground surface water sources.¹⁴ Additionally, reports of nitrate toxicosis in cattle following consumption of contaminated water delivered in tanks previously used to haul nitrogen-based fertilizers are concerning common. Campagnolo et al. (2002) described a case of nitrate toxicosis in 6 ruminants (4 cattle and 2 small ruminants) at a county fair after they consumed water that had been hauled in a fertilizer tank. The animals exhibited seizures and died shortly thereafter.⁷ Villar et al. (2003) described two additional cases of cattle deaths after cattle consumed hauled water contaminated with nitrogen-based fertilizer.⁸ The recent case submitted to the KVDL described above demonstrates that, despite repeated reports and longstanding warnings, the hazardous practice of watering cattle from tanks that could contain nitrate contamination from nitrogen-based fertilizers persists. Water testing potentially could have prevented the loss of animals in these situations. However, quantitative testing is often impractical due to multiple day lag time between sample collection to reporting when producers are hauling water under time-sensitive circumstances. Based on results from this study, some commercially available nitrate test strips provide a quick, affordable and user-friendly alternative for screening sources of livestock drinking water for nitrate contamination. These can be deployed on-farm and negate the need for shipping samples to analytical laboratories.

The results of this study demonstrated that test strip performance varies substantially by brand. Some brands performed well as screening tools for nitrate contamination, while others

underperformed. Therefore, careful selection of test strips is critical. This study focused exclusively on strips marketed for human drinking water, not those intended for swimming pools. Additionally, human drinking water standards set a nitrate limit of 10 ppm, whereas nitrate concentrations in water for cattle consumption are not typically concerning until they reach 100 ppm. It is essential to use a test capable of detecting nitrate levels at or above this threshold. Evaluation of diagnostic tests involves an assessment of the probability of false-positive and false-negative test results and the costs associated with each (Table 3). In the case of testing water for livestock, the cost of a false-positive nitrate test involves the loss of the water source, which may be substantial if another source for water or tank to haul water is not available. The cost of a false-negative result could be substantial if it results in intoxication and death loss within a cattle herd. Positive and negative predictive values intuitively describe the probability of false-positive and false negative-tests. For this study, positive predictive value is defined as the probability that the targeted nitrate concentration of the water sample truly is ≥ 100 ppm following a positive test result (i.e., the probability that the positive test sample is truly positive, and not a false positive). Conversely, negative predictive value is defined as the probability that the water sample nitrate concentration is < 100 ppm following a negative test result (i.e., the probability that the negative test sample is truly negative, and not a false negative). SJ wave, AssuTest, JNW Direct, and Varify all had positive predictive values greater than 90%, suggesting high confidence that when the test was positive, the water was also positive (few false positives), which is desirable for confirmatory tests when the cost of a false positive is high (e.g., loss of

Table 3: Diagnostic test measures, definitions and utility.

Measure	Definition	Utility
Sensitivity	Proportion of truly positive samples (denominator) that test positive (numerator).	This ability of a diagnostic test to detect positive samples correctly is a key characteristic of screening tests where a false negative test result is most “costly” and should be minimized.
Specificity	Proportion of truly negative samples (denominator) that test negative (numerator).	This ability of a diagnostic test to detect negative samples correctly is a key characteristic of confirmatory tests where a false positive test result is most “costly” and should be minimized.
Positive predictive value	Proportion of test positive samples (denominator) that are truly positive (numerator).	These diagnostic measures are indicators of the usefulness of a test when the true status of a sample is unknown (i.e., positive or negative). However, since these values are driven by the proportion of positive samples within the source population where the test is used (i.e., prevalence), predictive values can only be interpreted for that known prevalence.
Negative predictive value	Proportion of test negative samples (denominator) that are truly negative (numerator).	
Likelihood ratio for positive test results	The ratio of the probability of a positive test result among positive samples to the probability of a positive test result among negative samples. (True positive risk/ False positive risk; test sensitivity/[1-test specificity]).	Since likelihood ratios are not affected by the proportion of positive samples within a source population (prevalence), they are a better indicator of how likely a sample is to be positive given a specific test result. As a general rule, likelihood ratios for a positive test result above 10 and negative test result below 0.1 are considered good. A single test that can meet or exceed both likelihood ratio standards would be ideal, but in the case of a screening test with a goal of minimizing false-negative test results, a test with a low likelihood ratio for a negative test result would be preferable.
Likelihood ratio for negative test results	The ratio of the probability of a negative test result among positive samples to the probability of a negative test result among negative samples. (False negative risk/ True negative risk; [1-test sensitivity]/test specificity).	

water source). Only Bartovation had a negative predictive value > 90%, suggesting high confidence that when the test was negative the water was truly negative, which is desirable for screening tests where the cost of a false negative is high (e.g., cattle death loss).

While intuitively understandable, predictive values are dependent on the prevalence of positive samples. The predictive values reported here are for a 50% prevalence of truly positive samples. If the prevalence of nitrate-contaminated water sources is lower, negative predictive values would be expected to increase, and positive predictive values would decrease. Less intuitive measures of whether a test result is correct are likelihood ratios, which are not dependent on prevalence. For this study, the definition of the likelihood ratio is the ratio of the probability that the sample came from a positive vs. a negative water source. Based on results from this study, a positive result from an SJ wave test would be 31 times more likely to be from a nitrate-contaminated water source than a non-contaminated source (i.e., a positive test result is very unlikely to be from a negative water source). Conversely, a negative result from a Bartovation test would be 0.034 times as likely to be from a contaminated water source than a non-contaminated source (i.e., a negative test result is very unlikely to be from a positive water source). As a general rule, likelihood ratios above 10 (positive) and below 0.1 (negative) are considered good. Among the tests evaluated here, no test had both a

positive likelihood ratio above 10 and a negative likelihood ratio below 0.1. SJ wave and AssuTest had positive likelihood ratios above 10, and Bartovation had a negative likelihood ratio below 0.1. Test choice should be based on the scenario-specific risk and costs of a false positive or false negative result, because no single test provided high confidence in both positive and negative results. For the highest confidence, a sequential testing scheme could be implemented, where all water sources are screened for nitrate with a high-sensitivity test (e.g., Bartovation, 97.5% sensitivity), where the low-negative likelihood ratio makes false-negative test results unlikely, followed by retesting positive tanks with a high-specificity test (e.g., SJ wave, 97.5% specificity).

One important limitation of nitrate test strips is that they only detect nitrate and do not account for other potentially toxic compounds that are found in nitrogen-based fertilizers. The most commonly used fertilizers are composed of a mixture of ammonium nitrate and urea. These ingredients can lead to urea or ammonia toxicity, as well as nitrate toxicity, all of which can result in acute cattle deaths. These different toxicoses can also occur simultaneously within a single animal. A “safe” nitrate test result does not guarantee that a water source is free of other harmful contaminants. While this limitation is acknowledged, it was not addressed within the scope of this study. Producers must consider the specific chemical composition of any suspected fertilizer contamination before deeming a water source safe for cattle consumption.

Water testing laboratories report nitrate concentrations in multiple different units which are not equivalent. This inconsistency of unit reporting can be confusing for recipients of these reports to interpret. The molecular weight of the nitrate unit reported drives the interpretation and risk thresholds for the individual reports, and should not be applied to other results. For example, nitrate concentrations can be reported as either nitrate-nitrogen ($\text{NO}_3\text{-N}$) or nitrate ion (NO_3^-), with a conversion factor of 4.4 between the two, meaning the risk thresholds for each result are 4.4 times different; therefore, the absolute concentration values should be interpreted differently with regard to toxicity risk. The concentration of 100 ppm nitrate was used as the risk threshold for this project because it provides a safety factor below toxic concentrations. The evaluated colorimetric strips use a color gradient to estimate nitrate concentrations, where the color scale often increases in darkness with increasing nitrate concentrations. Without the precision of calibrated analytical techniques, setting risk thresholds at the point of acutely toxic concentrations is not ideal. Practically, we recommend that water samples with concentrations exceeding 100 ppm nitrate by colorimetric test strip undergo confirmatory testing with more precise analytical methods.

Although providing water to livestock that could be contaminated with nitrates is strongly discouraged, in water shortage emergency situations, nitrate test strips can serve as a useful tool to reduce the risk of catastrophic loss due to nitrate toxicosis. However, the accuracy and reliability of the test depend on selecting a test strip brand with the ability to detect dangerous concentrations. If a nitrate concentration of ≥ 100 ppm is detected when screening with test strips, it is highly recommended that the water undergoes further laboratory testing before being deemed safe for cattle consumption.

Endnotes

^a Thermo Electron, West Palm Beach, FL

^b SJ wave® 16-in-1 Water Test Kit, SJ Wave, Mount Waverley, VIC, Australia

^c Varify® 17-in-1 Complete Drinking Water Test Kit, Varify Inc., Coronado, CA

^d Bartovation® Nitrite, Nitrate Two Pad Test Strip, Bartovation, West Harrison, NY

^e JNW Direct® Drinking Water Test Kit 15 in 1, JNW Direct, Pinner, GB

^f AssuTest® 17-in-1 drinking water test kits, Hangzhou, China

^g MilliporeSigma™ MQuant™ Nitrate Test Strips, Burlington, MA

^h Inorganic Ventures, Christiansburg, VA

ⁱ Microsoft Excel, Microsoft Corporation, Redmond, WA

^j StataCorp. 2019. Stata: Release 16. Statistical Software. College Station, TX: StataCorp LLC

^k R version 4.4.1 2024 The R Foundation for Statistical Computing Platform

^l Rstudio 2025.05.0 Build 496 Posit Software, PBC

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Conflicts of Interest

The authors declare that this research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. No AI-assisted technologies were used in the generation of this manuscript.

Author contributions

BAH: methodology, validation, investigation, resources, data curation, writing (original draft, writing) review and editing, visualization, project administration, funding acquisition.

CJH: conceptualization, methodology, resources, data curation, writing (original draft, writing) review and editing, visualization, supervision, project administration.

SRC: resources, writing (original draft, writing) review and editing, visualization.

MWS: conceptualization, methodology, validation, formal analysis, data curation, writing (original draft, writing) review and editing, visualization, supervision.

MT: resources, writing – review and editing.

SAF: conceptualization, methodology, validation, resources, writing (original draft, writing) review and editing, visualization, supervision, project administration, funding acquisition.

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Appendix A: Nitrate data collection sheet.

Evaluator ID: 1

Test: A

Instructions: Follow manufacturer label instructions and record the result to the nearest numerical value in the corresponding dilution row.

Dilution ID	Result (Nitrate ppm)
a	
b	
c	
d	
e	
f	
g	
h	